

Aryakrishnan C R

Data Engineer | Cloud Data Engineering & Pipeline Automation

 aryakrishnancr@gmail.com

 +91 6238161638

 India

 LinkedIn

 GitHub

 my-portfolio-ten-pi-50.vercel.app/

 3 years experience

Professional experience

Accenture (Amex client)

Feb 2024 - Present · Bangalore

Data Engineering Analyst - Full Time

Nov 2025 - Present

- Architected an LLM-orchestrated PII detection service (FastAPI, Gemini/GPT) that traces column-level lineage through translated BigQuery SQL flows to auto-classify PII roles for every output table, eliminating manual PII tagging; used sqlglot-based dependency parsing and adaptive chunking to reliably scale to flows with 100+ SQL blocks.
- Built a schema-constrained LLM invocation layer with structured-output parsing, retry/backoff resilience, and custom SQL sanitization, producing full lineage-based reasoning (e.g., encryption/decryption tracing) for every PII classification decision.
- Involved in Project Ascend native cloud migration, transitioning System of Record (SOR) tables from legacy Cornerstone on-premise replication to native GCP Lumi Direct ingestion pipelines, eliminating legacy dependencies.
- Engineered multi-environment pipeline deployment across E1/E2/E3 using CDM-MDM converters, MDM APIs, and Apache Airflow DAGs (`lumi-sor-ingest`, `lumi-sor-load`) with WireSafe metadata encryption and Kafka-to-Pub/Sub real-time streaming.
- Developed an Ascend Data Validation Tool on BigQuery with a two-phase validation framework (checksum comparison + key-based joins) and hash/aggregation-based scalable validation, cutting manual effort 70% and runtime 50% vs. row-level processing.
- Built mismatch analysis on primary/composite keys, generating match/mismatch metrics with 95% accuracy across edge cases like missing keys, floating-point precision, PII data, and nested/struct data extending coverage 40%.
- Automated workflows with Apache Airflow, enabling hourly validation (down from 1 day) and generating reports in BigQuery/GCS, cutting manual reporting effort by 60%.

Data Engineering Associate

Feb 2024 - Nov 2025

- Developed automated BigQuery-to-BigQuery Data Validation Tool (DVT) checks across 7-day ingestion windows to validate schema drift, row counts, PII/decryption accuracy, and millisecond timestamp precision.
- Developed a data validation tool to automate data comparison between on-premises and GCP BigQuery environments, increasing validation efficiency by 90%; automated creation and execution of BigQuery and Hive queries with Pub/Sub integration, reducing manual processing by 30 hours/month.
- Built a cross-environment table comparison application for schema validation and data integrity checks, improving data ingestion efficiency by 80%, using Apache Airflow and GCP services to design scalable, real-time ingestion pipelines.
- Established operational SOPs and automated retry mechanisms via Refresh and Stargate APIs to resolve pipeline metadata and ingestion failures across production cutover and hypercare phases.

Achievements & Certifications

- Awarded the Amex Bright Beginner Award for embodying Accenture core values on the American Express account.
- Awarded the Accenture(Amex) Star Award
- Received the Delivery Excellence Award for building scalable data pipelines using PySpark and Airflow.
- Google Cloud Digital Leader (Jul 2024 - Jul 2027)

Profile

Data Engineer specializing in Cloud Data Engineering & Pipeline Automation with expertise in building LLM-orchestrated pipelines, native cloud migration architectures, scalable data validation frameworks, and automated cross-environment ingestion systems on GCP.

Core expertise

- PythonSQLHTML/CSSJava
- Google Cloud Platform (GCP)Apache Airflow
- FastAPI

Skills

- HadoopPySparkHiveHDFS
- DatabricksSpark Structured Streaming
- DataprocComposerCloud Storage (GCS)
- Pub/SubBigQueryKafka
- LLM-based pipelines (Gemini, GPT)
- Schema-constrained generation
- Prompt engineering
- SQL lineage parsing (sqlglot)

Approach

- Cloud Data EngineeringPipeline Automation
- Agentic AI / LLM Engineering
- SQL Lineage Parsing
- Data Validation Frameworks
- End-to-End Database Migration Lifecycle
- Zero-Downtime Cutover & Hypercare Support

Tools

- GitGitHub ActionsDatabricks
- Apache AirflowGoogle Cloud Platform
- FastAPIJiraBigQuery DVT

Education

Bachelor of Technology
Computer Science
Amal Jyothi College of Engineering / India
June 2019 - April 2023 / 8.0 CGPA